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Chemoenzymatic Synthesis of Rhamnolipid-Modified Exendin-4: A Dual Approach for Long-Acting Glycemic Control through Endogenous Antibody Recruitment and Albumin Interaction

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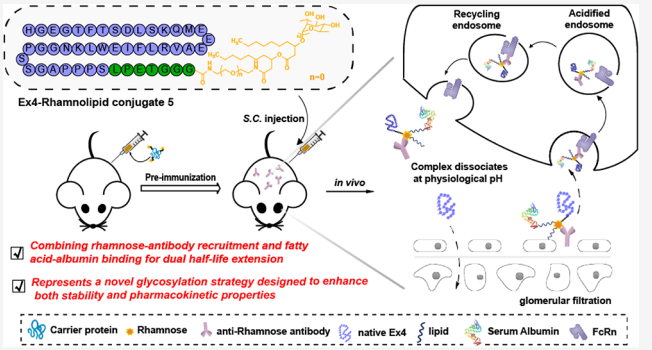
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ABSTRACT: Effective management of Type 2 diabetes requires therapies that balance potency with sustained action. Glycosylation of glucagon-based drugs improves stability and reduces aggregation but does not significantly enhance pharmacokinetics. This study develops a novel glycosylation strategy—a rhamnolipid conjugation platform—to address this challenge. Using sortase A-mediated enzymatic ligation, we synthesized Ex4−rhamnolipid bioconjugates that prolong action through a dual mechanism: carbohydrate-mediated antibody engagement and lipophilic albumin binding. These conjugates potently activated GLP-1 receptors (EC_{50} = 0.14−0.21 nM) and elicited rapid blood glucose reduction in diabetic rodent models. In immunized subjects, glucose-lowering effects persisted for 24 h, reflecting a 6.7-fold improvement in circulation half-life. In extended-duration studies, once-daily administration of the lead conjugate achieved glycemic control and pancreatic β-cell protection comparable to semaglutide without adverse effects. This methodology capitalizes on natural immune components and plasma protein interactions to address the challenges of peptide-based treatments, offering a promising platform for sustained-release antidiabetic drugs.



INTRODUCTION

Type 2 diabetes mellitus (T2DM) represents a multifaceted endocrine disease marked by disturbances in the metabolism of carbohydrates, fats, and proteins, primarily caused by either deficient insulin production, cellular insulin insensitivity, or a combination of these factors.[1] Recent epidemiological data indicate that more than 589 million individuals globally were affected by this condition in 2025, with numerous patients developing dangerous associated conditions, including vision impairment, lower extremity complications, metabolic-associated fatty liver disease, and excessive weight gain that may threaten survival. Such statistics emphasize the critical demand for innovative treatment options and breakthrough medications targeting diabetes.[2,3] Among current therapeutic options, incretin-mimicking peptides, including GLP-1 and its analog Ex4, have gained prominence in diabetes care owing to their unique ability to stimulate insulin release in a glucose-concentration-dependent manner, while presenting negligible risks of dangerously low blood sugar, promoting body weight reduction, and potentially offering protection against heart disease.[4] Nevertheless, the medical use of these peptide drugs faces substantial obstacles due to their inherent molecular vulnerabilities—swift enzymatic breakdown and efficient kidney filtration—which dramatically shorten their biological persistence, requiring repeated dosing at elevated quantities that significantly compromise treatment adherence.[5,6] Consequently, developing a new strategy to enhance the incretin-mimicking peptide pharmacokinetics has become a paramount objective in diabetes research.

Current strategies to improve the pharmacokinetics of peptide drugs include PEGylation, lipidation, Fc fusion, and glycosylation. Among these, lipidation—exemplified by clinical peptides such as liraglutide, semaglutide, and tirzepatide—enhances albumin binding via fatty acid chains, forming stable drug-albumin complexes that reduce renal clearance.[7−9] Similarly, Fc fusion proteins, such as dulaglutide, achieve a prolonged half-life (up to 50-fold longer than native GLP-1) through FcRn-mediated recycling, enabling weekly dosing. While these approaches effectively extend circulation time,

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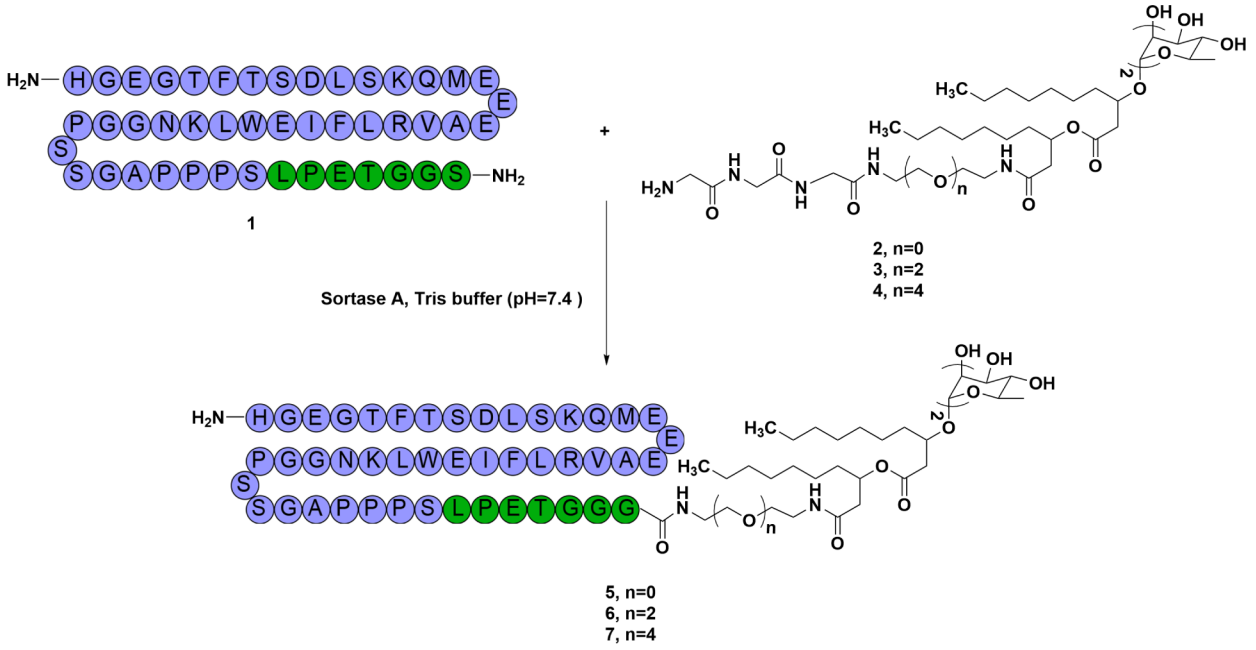


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Scheme 1. Chemoenzymatic Synthesis of Ex4-Rhamnolipid Conjugates 5−7



they face limitations such as potential immunogenicity, manufacturing complexity, and product heterogeneity, which may hinder clinical translation.[10−12] In contrast, glycosylation offers a distinct set of advantages as a post-translational modification strategy. By incorporating structurally defined carbohydrates—including *N*- and O-linked oligosaccharides—into peptides such as insulin and GLP-1, glycosylation preserves biological activity while significantly improving solubility and suppressing fibrillation.[13−16] These properties make glycosylation particularly valuable for enhancing drug stability and reducing aggregation-related challenges. However, unlike lipidation or Fc fusion, glycosylation does not substantially improve the pharmacokinetic profile of glucagon-based drugs *in vivo*, limiting its utility for applications requiring prolonged systemic exposure. Thus, developing novel glycosylation strategies for glucagon-based drugs-ones that retain biological activity while improving pharmacokinetics—could enhance their therapeutic potential for metabolic disorders.

Endogenous antibodies are naturally generated immunoglobulins that recognize various haptens, such as nitroarene compounds and carbohydrate epitopes, including α-Gal, Forssman disaccharide, and L-rhamnose (Rha).[17−20] A peptide conjugate containing dinitrophenylamine (DNP) is able to bind naturally occurring antibodies, effectively prolonging Ex4's biological persistence.[21] Rhamnolipids are glycolipid biosurfactants featuring a distinctive structure composed of one or two Rha moieties as the hydrophilic headgroup and two fatty acid chains as the hydrophobic tail.[22,23] This molecular architecture positions rhamnolipid-modified GLP-1 peptides as promising bifunctional agents capable of both Rha-mediated antibody binding and fatty-acid-driven albumin interaction, thereby overcoming the pharmacokinetic challenges associated with native peptides. In this study, we designed and synthesized a series of C-terminal rhamnolipid-modified Ex4 analogues incorporating polyethylene glycol (PEG) spacers of varying lengths via sortase A-mediated ligation (SML) (Scheme 1). This novel glycosylation strategy capitalizes on: (1) the binding between the Rha hapten and endogenous antibodies and (2) the natural albumin-binding capacity of the lipid tail, collectively enabling a dual mechanism for optimizing Ex4's pharmacokinetic profile and extending its half-life. We systematically evaluated the conjugates' *in vitro* GLP-1R activation potential, acute and chronic glucose-lowering efficacy *in vivo*, and pharmacokinetic properties. Furthermore, we characterized the long-term therapeutic effects of our lead compound through a comprehensive seven-week study in db/db mice.

RESULTS

Synthesis of Ex4-Rhamnolipid Conjugates 5−7 via SrtA-Mediated Ligation. To prepare Ex4-Rhamnolipid conjugates via SML,[24,25] we first synthesized oligoglycine-functionalized rhamnolipid derivatives (**2**−**4**) bearing PEG spacers of varying lengths (n = 0, 2, 4) following our established protocol.[26] The Ex4-LPETGGS (**1**) peptide was synthesized using standard solid-phase peptide synthesis (SPPS) techniques.[21] Conjugation reactions were carried out by incubating peptide **1** with each rhamnolipid derivative (**2**-**4**) in the presence of sortase A (Scheme 1). Reaction progress was monitored by reversed-phase HPLC (RP-HPLC), which revealed efficient conversions with yields of 90% (**5**), 88% (**6**), and 90% (**7**) (Figure 1a−c). Electrospray ionization mass spectrometry (ESI-MS) confirmed the successful formation of target conjugates **5**−**7**. After scaling up the reactions, we obtained milligram quantities of each conjugate (**5**: 3.5 mg, 62% yield; **6**: 3.7 mg, 65% yield; **7**: 3.5 mg, 60% yield) following semipreparative HPLC purification. Analytical HPLC demonstrated >95% purity for all final compounds (Figures S3−S5), meeting the requirements for subsequent biological evaluation.

GLP-1 Receptor Activation Potency. The activation of GLP-1R by its agonists triggers cyclic adenosine monophosphate (cAMP) release as a secondary messenger, initiating downstream signaling cascades that ultimately promote insulin secretion and blood glucose regulation.[27−29] To evaluate the

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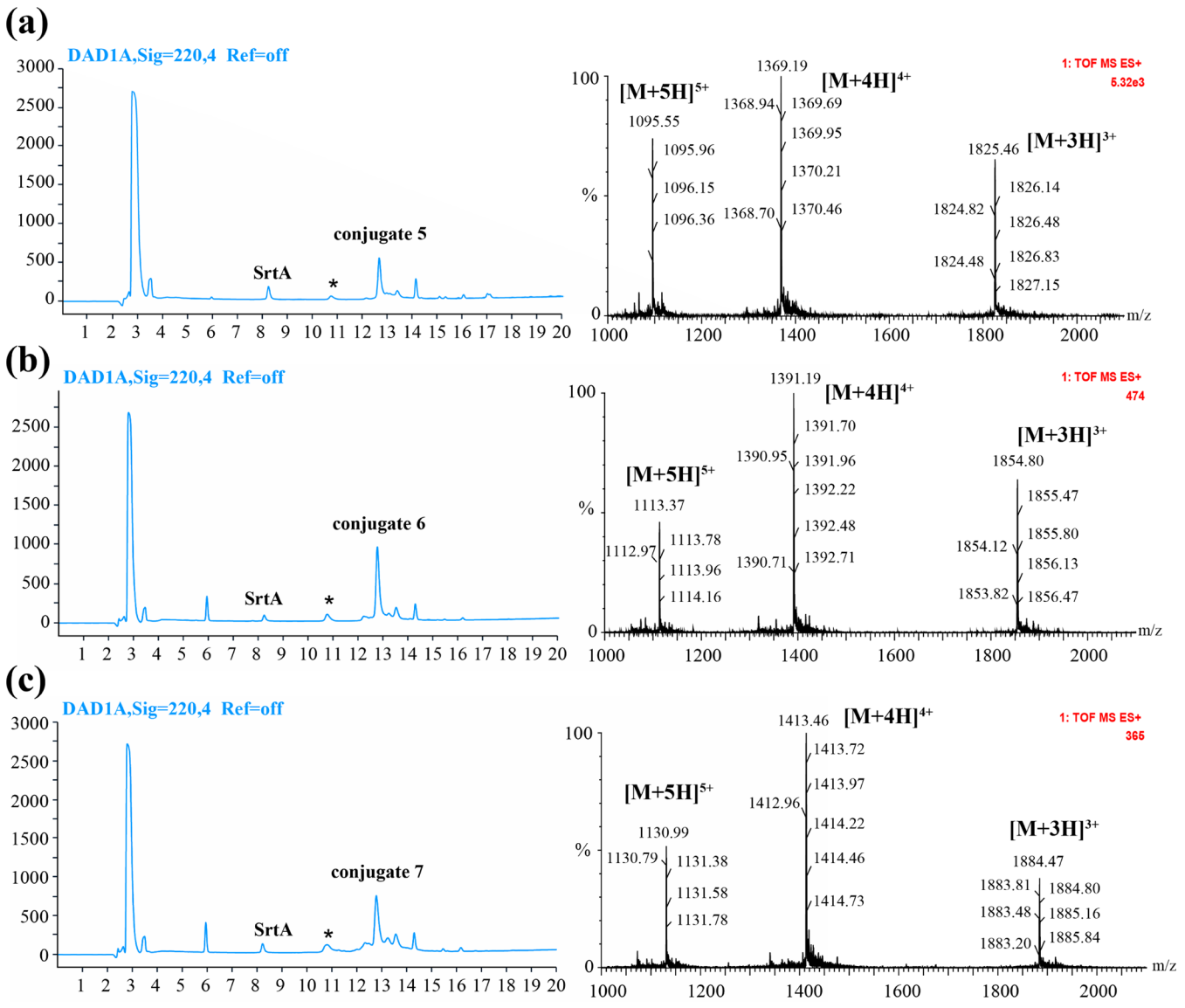


Figure 1. HPLC profile of SrtA-mediated ligation and MS spectrum of Ex4-Rhamnolipid conjugates **5** (a), **6** (b), and **7** (c). HPLC conditions: gradient 10−90% CH_3CN/H_2O over 20 min at a flow rate of 1 mL/min, monitored at 220 nm (*denoting unreacted Ex4 analogue **1** or its hydrolysis product).

pharmacological activity of Ex4-Rhamnolipid conjugates **5**−**7**, we performed cAMP accumulation assays in HEK-293 cells stably expressing human GLP-1R. As shown in Figure 2, all conjugates demonstrated concentration-dependent cAMP production following 20 min incubations, though with slightly reduced potency compared to native Ex4. Importantly, all conjugates maintained subnanomolar EC_{50} values (**5**: 0.14 nM; **6**: 0.19 nM; **7**: 0.21 nM), indicating preserved high receptor affinity. Notably, the cAMP stimulation capacity showed a modest PEG-length dependence, with conjugate **5** (shortest linker) exhibiting the most favorable profile among the series.

Acute Glucose-Lowering Effect of Ex4-Rhamnolipid Conjugates on db/db Mice. To evaluate the acute glucose-lowering efficacy of Ex4-Rhamnolipid conjugates *in vivo*, we conducted intraperitoneal glucose tolerance tests (IPGTT) in $Lepr^{db}$ mutant C57BLKS/J (db/db) mice, a well-established type 2 diabetes model. Following a single intraperitoneal dose (25 nmol/kg) of PBS, native Ex4, or conjugates **5**−**7**, mice received a glucose challenge (1 g/kg, i.p.). As shown in Figure 3a, PBS-treated mice displayed rapid hyperglycemia, peaking at 23.4 mmol/L within 30 min but failing to return to baseline

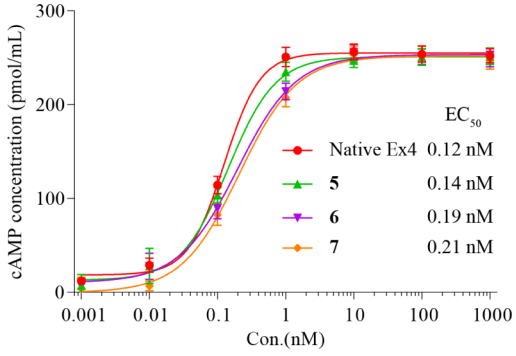


Figure 2. Assessment of the hGLP-1R agonist potency for Ex4-Rhamnolipid conjugates **5**−**7**. The data were obtained using a competitive cAMP assay with hGLP-1R stably transfected HEK293 cells, with plots representing mean ± SEM (n = 3).

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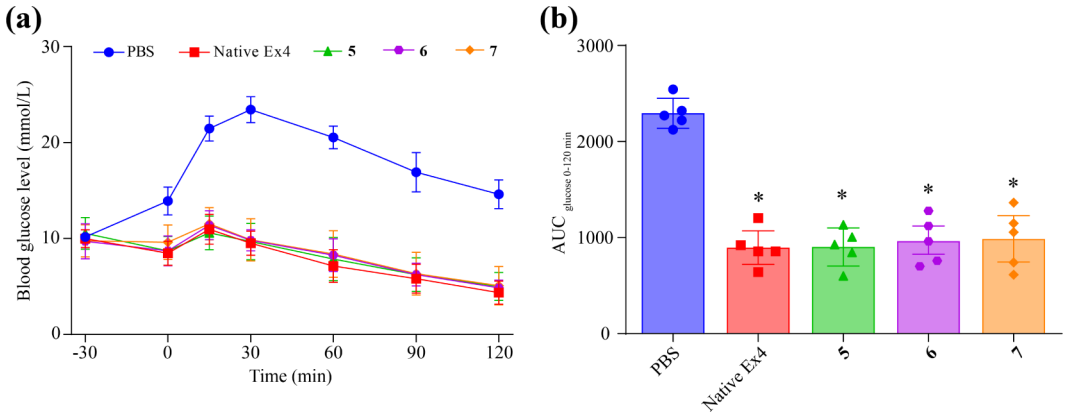


Figure 3. Effects of PBS, native Ex4, and Ex4-Rhamnolipid conjugates **5**−**7** (25 nmol/kg) on IPGTT in db/db diabetic mice. (a) The blood glucose level-time response curve of each group. (b) Quantitative analysis of the acute hypoglycemic effect by $AUC_{glucose\ 0-120\ min}$. The results were expressed as mean ± SEM (n = 5), *p < 0.05 vs PBS.

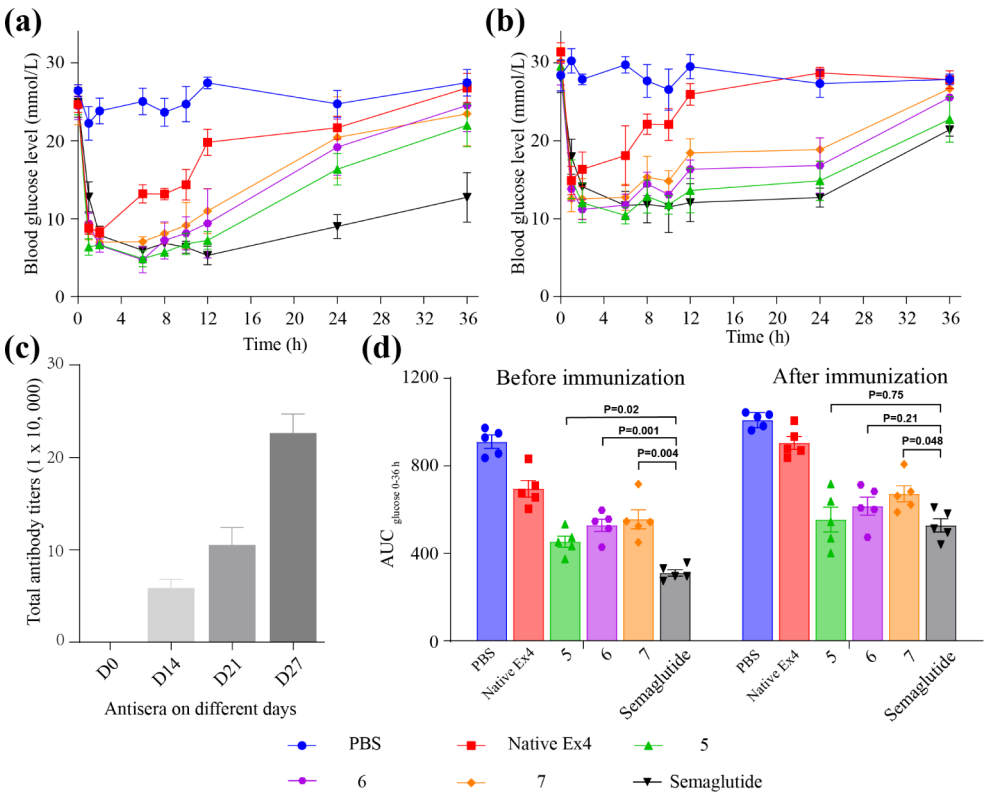


Figure 4. Long-lasting hypoglycemic effects of Ex4-Rhamnolipid conjugates before and after preimmunization in db/db mice. (a) Without or (b) with anti-Rha antibody in serum, the glucose-lowering profile of PBS, native Ex4, Ex4-Rhamnolipid conjugate **5**−**7**, and semaglutide groups (*s.c.*, 25 nmol/kg) in a db/db mouse model. (c) Assessment of the anti-Rha antibody titers of the serum collected on different days in the db/db mice after immunization with the Rha-OVA conjugate (n = 25). (d) The calculated blood glucose AUC values from 0 to 36 h in PBS, native Ex4, Ex4-Rhamnolipid conjugate **5**−**7** and semaglutide groups (n = 5). Results were depicted as means ± SEM, p < 0.05 represents statistically significant differences.

within 2 h, consistent with their impaired glucose regulation. In contrast, all treatment groups showed attenuated glycemic responses, with peak glucose levels of only 10.9 (native Ex4), 10.6 (**5**), 11.4 (**6**), and 11.5 mmol/L (7) at 15 min, followed by complete normalization within 60 min. Quantitative analysis revealed significant reductions in $AUC_{glucose\ 0-120\ min}$ values versus PBS controls (60.5% for Ex4 vs 60.2%, 57.5%, and 56.6% for **5**−**7**, respectively; *p < 0.05), with no statistically significant differences between native Ex4 and conjugates (Figure 3b). These results demonstrate that C-terminal rhamnolipid modification preserves the potent glucose-regulating activity of Ex4 without compromising its insulin secretion-promoting biological activity.

Long-Term Hypoglycemic Activity of Ex4-Rhamnolipid Conjugates in db/db Mice. To validate our hypothesis that rhamnolipid modification extends Ex4's half-life through a dual mechanism, we evaluated the hypoglycemic duration of Ex4-Rhamnolipid conjugates in db/db mice with and without anti-Rha antibodies. In nonimmunized mice (Figure 4a), all treatment groups, except PBS, achieved normoglycemia (<15

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mmol/L) within 1 h. Notably, conjugates **5**−**7** maintained significantly lower glucose levels than native Ex4 at 6, 8, and 12 h ($p < 0.05$), suggesting that albumin binding alone provides substantial prolongation of activity. To assess the antibody-dependent mechanism, we immunized mice with Rha-OVA to generate high anti-Rha antibody titers (Figure 4c). In these immunized mice (Figure 4b), conjugates **5**−**7** showed enhanced glucose control versus that of native Ex4, with effects persisting through 24 h. Their glycemic profiles closely resembled those of semaglutide-treated controls. Quantitative analysis revealed dramatic improvements in $HGD\%_{0-36\ h}$ values postimmunization (**5**: 46.0% → 5.0%; **6**: 70.3% → 16.6%; **7**: 79.2% → 27.4%; Figure 4d and Table 1), confirming the critical role of anti-Rha antibodies in the dual mechanism. Consistent with our structure−activity relationship analysis, conjugate **5**, which incorporates the shortest linker, exhibited superior performance, paralleling the sustained efficacy of semaglutide in immunized mice. These results demonstrate that (1) fatty acid-albumin interaction provides baseline prolongation and (2) anti-Rha antibody binding synergistically enhances duration, with linker length critically modulating this effect.

Table 1. *In vivo* $AUC_{glucose\ 0-36\ h}$ Values of Native Ex4, Ex4-Rhamnolipid Conjugates, and Semaglutide in the Hypoglycemic Duration Test

Compounds	$AUC_{glucose\ 0-36\ h}$[a]			
	Before immunization	$HGD\%_{0-36\ h}$[b]	After immunization	$HGD\%_{0-36\ h}$[b]
PBS		\	1014 ± 22	\
Native Ex4	695 ± 57	\		\
5	453 ± 34	46.0%	554 ± 33	5.0%
6	528 ± 29	70.3%	615 ± 59	16.6%
7	556 ± 37	79.2%	672 ± 49	27.4%
semaglutide	310 ± 27	\	528 ± 43	\

[a]The $AUC_{glucose\ 0-36\ h}$ values were calculated from the results of the hypoglycemic duration test ($n = 5$). Results were reported as means ± SEM. [b]The hypoglycemic degree ($HGD\%_{0-time}$) of samples, compared to the positive control of semaglutide, was defined as follows: $HGD\%_{0-36\ h} = [(AUC_{semaglutide,\ 0-36\ h} - AU_{test,\ 0-36\ h})/AUC_{semaglutide,\ 0-36\ h}] \times 100\%$.

***In Vivo* Pharmacokinetic Study in Balb/C Mice Preimmunized With or Without Rha-OVA.** To validate the dual mechanism of half-life extension mediated by rhamnolipid modification, we conducted a comparative pharmacokinetic study of conjugate 5 (75 nmol/kg, i.v.) in Balb/c mice, both before and after preimmunization with Rha-OVA.[30,31] In nonimmunized mice (Figure 5a), the plasma concentration−time profile of native Ex4 exhibited rapid clearance ($T_{1/2}$ = 0.32 ± 0.05 h), consistent with its unmodified peptide structure. Under the same conditions, conjugate 5 demonstrated a significantly prolonged circulation time, with a 3.5-fold increase in elimination half-life ($T_{1/2}$ = 1.18 ± 0.12 h). This result confirms that the albumin-binding capacity of the lipid tail alone provides substantial pharmacokinetic enhancement, even in the absence of an antirhamnose antibody interaction. We next assessed the contribution of the antibody-dependent mechanism in mice immunized with Rha-OVA to generate high anti-Rha antibody titers. In this model (Figure 5b), native Ex4 clearance remained rapid and unchanged ($T_{1/2}$= 0.34 ± 0.07 h), confirming that immunization itself does not alter the pharmacokinetics of the native peptide. Strikingly, the half-life of conjugate 5 in immunized mice was 6.7-fold longer than that of native Ex4 ($T_{1/2}$= 2.27 ± 0.17 h). Crucially, this was also nearly twice as long as its half-life in nonimmunized mice. These pharmacokinetic data are consistent with the observed enhancements in long-acting hypoglycemic activity, conclusively demonstrating that rhamnolipid modification optimally extends the plasma half-life of Ex4 through two complementary pathways.

Chronic Treatment Effects of Ex4-Rhamnolipid Conjugate 5. Among the rhamnolipid-modified Ex4 conjugates, compound **5** demonstrated optimal pharmacokinetic properties and sustained hypoglycemic activity, leading to its selection for long-term evaluation in immunized db/db mice over a seven-week period. Based on pharmacokinetic profiling, we established a once-daily dosing regimen for conjugate **5**, compared to twice-daily administration for native Ex4. Semaglutide served as the positive control in these chronic treatment studies.

Effect on Body Weight, Food Intake, Water Consumption, and Fasting Blood Glucose Levels. During the seven-week treatment period, we systematically tracked body weight, food/water consumption, and fasting blood glucose (FBG) levels in all experimental groups. As shown in Figure 6, all active treatments (native Ex4, conjugate **5**, and semaglutide) demonstrated significant effects on weight management. By

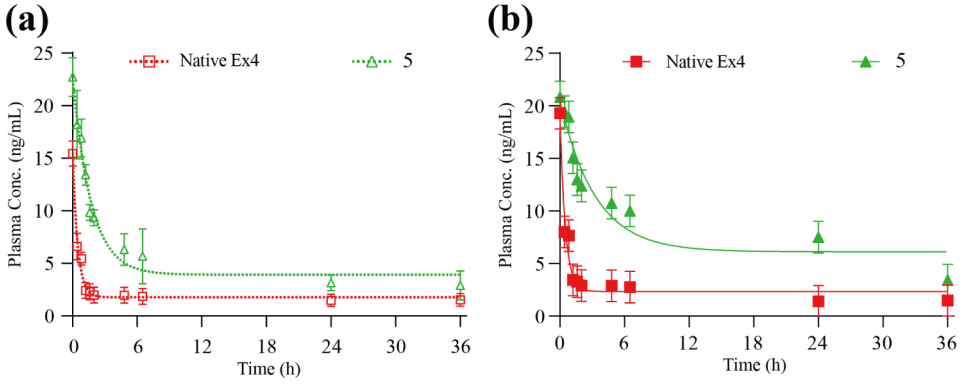


Figure 5. Pharmacokinetic evaluation of native Ex4 and Ex4-Rhamnolipid conjugate **5** in Balb/c mice. Plasma concentration−time profiles were measured (a) before and (b) after preimmunization with Rha-OVA. Mice received a single intravenous dose (75 nmol/kg), and blood samples were collected at indicated time points for exendin-4 quantification by ELISA. Data are presented as mean ± SEM ($n = 3$).

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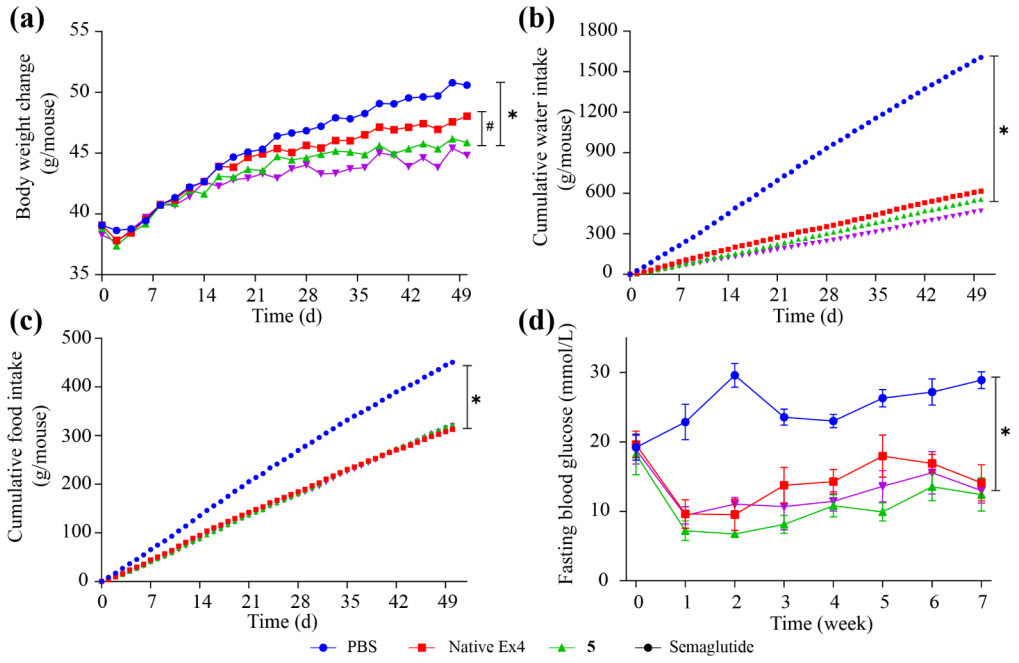


Figure 6. Effects on (a) body weight, (b) water intake, (c) food intake, and (d) fasting blood glucose levels (8 h fasting) in preimmunized db/db mice during long-term treatment experiments by subcutaneous injections of PBS, native Ex4 (twice daily), conjugate **5** (once daily), and semaglutide (once daily). The results are depicted as the mean ± SEM (n = 5), *p < 0.05 vs PBS and #p < 0.05 vs native Ex4.

study completion, conjugate **5** and semaglutide showed superior weight gain suppression (18.1% and 17.0%, respectively) compared with both PBS controls (29.4%) and native Ex4-treated mice (23.0%). Additionally, these therapeutic interventions consistently reduced food and water intake while maintaining significantly lower FBG levels (8-h fasted state) relative to PBS controls throughout the study duration.

Effect on Glucose Tolerance, HbA1c, Lipid Metabolism, and Toxicity Indicators. To evaluate treatment efficacy, we performed an IPGTT after the 7-week regimen. Both native Ex4 and conjugate **5** significantly improved glucose tolerance versus PBS controls (Figure 7a), as evidenced by reduced $AUC_{glucose\ 0-120\ min}$ values. Notably, conjugate **5** achieved comparable glucose tolerance improvement to semaglutide ($p > 0.05$). The therapeutic effects were further confirmed by HbA1c reduction, with conjugate **5** demonstrating a significant 3.0% decrease versus PBS controls and performance equivalent to that of semaglutide (Figure 7b,c). Lipid profiling revealed that conjugate **5**, like semaglutide and native Ex4, significantly lowered plasma TG and LDL-C levels versus PBS controls (*p < 0.05, Figure 7d−g), while HDL-C and TC remained unchanged across groups. The findings highlight conjugate **5**'s ability to modulate diabetic dyslipidemia with an efficacy level comparable to that of the clinical standard, semaglutide. Safety assessments showed no significant differences in hepatic (AST, ALT) or renal (Crea, Urea) toxicity markers between conjugate **5**- and PBS-treated mice (Figure S14), confirming the favorable safety profile of chronic conjugate **5** administration.

Histological Analyses and Immunohistochemistry. Histopathological evaluation of pancreatic tissues following chronic treatment revealed significant therapeutic benefits. H&E staining demonstrated that conjugate **5** treatment preserved pancreatic architecture, with well-defined islets and acini showing uniform distribution, a marked improvement over the disorganized morphology observed in PBS controls (Figure S15). Immunohistochemical analysis further revealed expanded insulin-positive β-cell areas (brown staining) in both conjugate **5**- and semaglutide-treated groups compared to the PBS controls. These findings highlight improvements in both pancreatic morphology and β-cell function in diabetic mice.

DISCUSSION AND CONCLUSIONS

Therapeutic peptides have emerged as a transformative class of pharmaceutical agents, with over 80 approved drugs currently addressing diverse clinical needs, ranging from metabolic disorders to oncology.[32,33] Particularly notable is the emergence of GLP-1-derived therapeutics, which account for 33 of the noninsulin peptide drugs approved since 2000.[34] However, the therapeutic potential of these peptides has been historically constrained by their rapid systemic clearance, necessitating innovative strategies to enhance their pharmacokinetic properties without compromising biological activity. Although current glycosylation strategies for glucagon-based drugs effectively enhance stability and reduce aggregation, they do not substantially improve pharmacokinetics *in vivo*—unlike lipidation or Fc fusion. This limitation restricts their practical utility in clinical applications. We address this critical need by developing a novel glycosylation strategy, the rhamnolipid conjugation platform, that simultaneously leverages endogenous antibody recruitment and albumin binding, thereby overcoming the challenges of immunogenicity and renal clearance that have hindered previous peptide drug development efforts.

Using SML, we efficiently synthesized three Ex4-Rhamnolipid conjugates (**5**−**7**) with varying PEG linker lengths (n = 0, 2, 4) in high yields (~90%). Importantly, C-terminal modification preserved receptor activation capacity (EC_{50} = 0.14−0.21 nM), as the rhamnolipid moiety was distally positioned from the crucial *N*-terminal activation motif. This structural preservation translated to maintained acute glucose-

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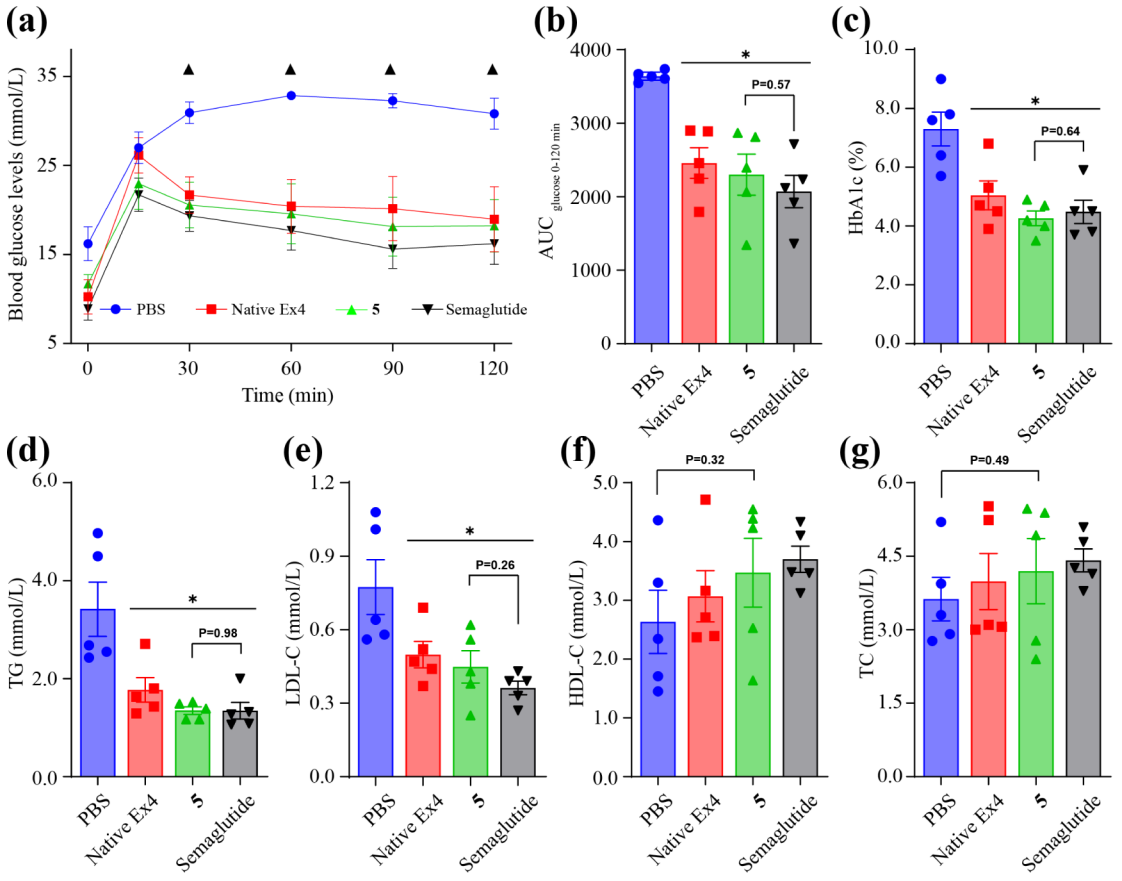


Figure 7. Chronic treatment effects of PBS, native Ex4, conjugate **5**, and semaglutide on glucose tolerance, HbA1c, and lipid metabolism indicators in preimmunized db/db mice. (a) Blood glucose excursion during IPGTT performed at the end of chronic treatment. (b) AUC of IPGTT for 0−120 min after the administration of glucose (dosage of 0.5 g/kg). (c) HbA1c levels, (d) plasma TG, (e) LDL-C, (f) HDL-C, and (g) TC levels measured after the sacrifice of animals. The results are depicted as the mean ± SEM (n = 5), *p < 0.05 vs the PBS group. (▲) Note: Since the range of the glucometer is 1−33.3 mmol/L, when the reading of the glucometer is displayed as "HI," maximal values of 33.3 mmol/L were substituted to enable statistical analysis.

lowering efficacy *in vivo*, with all conjugates showing comparable $AUC_{glucose\ 0-120\ min}$ values to native Ex4. Hypoglycemic duration testing revealed that Ex4-Rhamnolipid conjugates maintained superior glucose-lowering effects compared to native Ex4 in nonimmunized diabetic mice. This initial prolongation suggests substantial pharmacokinetic extension mediated by albumin binding, which is supported by the confirmed *in vitro* binding affinity of conjugate **5** for human serum albumin (K_d = 5.8 μM) (Figure S11). Notably, immunized mice with high anti-Rha antibody titers demonstrated a dramatically enhanced duration of action. This antibody-dependent enhancement is evidenced by significant improvements in $HGD\%_{0-36\ h}$ values postimmunization (**5**: 46.0% → 5.0%; **6**: 70.3% → 16.6%; **7**: 79.2% → 27.4%) and is corroborated by the high in vitro binding affinity of conjugate **5** for anti-Rha antibodies (Figure S12). Remarkably, conjugate **5** achieved hypoglycemic duration comparable to semaglutide, as demonstrated in multiple IPGTT experiments (Figure S13), despite its modest 6.7-fold half-life extension versus native Ex4. This exceptional result demonstrates that conjugate 5 provides substantial islet protective effects comparable to semaglutide. Its preserved performance is particularly noteworthy given semaglutide's established 165 h half-life in humans. Further pharmacokinetic evaluation in higher species (e.g., nonhuman primates) would provide valuable translational insights into this promising dual-mechanism approach. To evaluate long-term therapeutic efficacy, we conducted a seven-week chronic treatment study in immunized db/db mice using Ex4-rhamnolipid conjugate **5**. Once-daily administration of conjugate **5** demonstrated superior metabolic benefits compared to the twice-daily administration of native Ex4, including significantly improved control of water intake and body weight gain. Crucially, conjugate **5** treatment for 2 months produced substantial improvements in key diabetic parameters: reduced HbA1c levels, enhanced glucose tolerance, and normalized lipid profiles-effects that matched those achieved by once-daily semaglutide treatment. Histopathological examination revealed conjugate **5**'s protective effects on pancreatic islet morphology and function, comparable to semaglutide. Importantly, comprehensive toxicity assessments showed no adverse effects from chronic conjugate **5** administration.

In summary, we successfully developed a series of C-terminally rhamnolipid-modified GLP-1 analogues (**5**−**7**) through efficient chemoenzymatic synthesis. These conjugates maintained native Ex4-like potency in GLP-1R activation and acute glucose-lowering activity while achieving significantly prolonged half-lives through dual mechanisms: rhamnose-antibody binding and fatty acid-albumin interaction. Chronic studies demonstrated that lead conjugate **5**, administered once daily, matches the therapeutic efficacy of semaglutide across multiple metabolic parameters. This rhamnolipid modification

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strategy represents a versatile platform for developing long-acting peptide therapeutics. Given the growing prominence of peptide therapeutics, our approach offers significant potential for clinical translation across multiple therapeutic areas.[33]

EXPERIMENTAL SECTION

Materials. Fmoc-protected amino acids, Rink amide MBHA resin, 2-(1H-benzotriazole-1-yl)-1,1,3,3-tetramethyluronium tetrafluoroborate (TBTU), and 1-hydroxybenzotriazole (HOBt) were obtained from GL Biochem (Shanghai, China). Trifluoroacetic acid (TFA), triisopropylsilane (TIPS), and diisopropylethylamine (DIPEA) were purchased from J&K Chemical Technology Co., Ltd. (Beijing, China). Dulbecco's Modified Eagle Medium (DMEM, Cat. no. SH30243.01), fetal bovine serum (FBS, Cat. no. SH30084.03), penicillin–streptomycin solution (Cat. no. SV30010), and 0.25% Trypsin-EDTA (Phenol Red, Cat. no. SH30042.01) were sourced from Cytiva (Washington, US). The HbA1c assay kit (Cat. no. 044618002), TC assay kit (Cat. no. 141620003), LDL-C assay kit (Cat. no. 142019006), HDL-C assay kit (Cat. no. 142119010), TG assay kit (Cat. no. 141719008), urea assay kit (Cat. no. 141318005), Crea assay kit (Cat. no. 141119010), AST assay kit (Cat. no. 140218005), and ALT assay kit (Cat. no. 140120005) were acquired from Shenzhen Mindray Bio-Medical Electronics Co., Ltd. (Shenzhen, China). Additionally, HRP-labeled goat antimouse IgG antibody (Cat. no. A0216), HRP-labeled goat antirabbit IgG antibody (Cat. no. A0208), and TMB solution (for ELISA, Cat. no. P0209) were sourced from Beyotime (Shanghai, China). Other chemical reagents, unless otherwise specified, were obtained from commercial sources and used without further purification.

Microwave-Assisted Solid-Phase Peptide Synthesis. All peptides were synthesized using a Liberty Blue automated microwave peptide synthesizer (CEM Corporation, USA) under microwave conditions, following established protocols for 9-fluorenylmethoxycarbonyl (Fmoc)-based solid-phase peptide synthesis (SPPS), as previously described.[27] The standard microwave coupling conditions involved heating at 75 °C and 170 W for 15 s, followed by heating at 90 °C and 30 W for 110 s with the addition of amino acids (0.2 M), DIC (0.25 M), and Oxyma (0.5 M) in DMF. The standard microwave deprotection conditions involved heating at 75 °C and 155 W for 15 s, followed by heating at 90 °C and 30 W for 50 s with the addition of 20% piperidine in DMF. After the completion of synthesis, the resin was treated with a mixture of TFA/H_2O/TIPS (95:2.5:2.5, v/v/v) for 2 h at room temperature to cleave it. The resulting crude peptides were then centrifuged and triturated with cold ether to obtain a suspension, followed by semipreparative HPLC purification using an XBridge Prep C18 column (250 × 10 mm, 5 μm, 130 Å). The molecular weights of native Ex4 (HGEGTFTSDLSKQMEEEAVRLFIEWLKNGGPSSGAPPPS) and its analogues **1** (HGEGTFTSDLSKQMEEEAVRLFIEWLKNGGPSSGAPPPSLPETGGS) were determined using matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI TOF-MS, ultrafleXtreme, Bruker Daltonics Inc., USA) or electrospray ionization mass spectrometry (MALDI SYNAPT mass spectrometer, Waters, USA). The purity of native Ex4 and its analogues **1** was determined using analytical HPLC (Agilent 1260 Infinity) with a Diamonsil C18 column (250 × 4.6 mm, 3.5 μm, 100 Å), which confirmed that all peptides had a minimum purity level of ≥95% based on their UV absorption at 220 nm. The results are presented in Figures S1 and S2.

Synthesis of Rha Derivatives 2–4. Small rhamnolipid R3 derivatives **2–4** were synthesized following our previous procedures.[26]

SrtA-Mediated Ligation Reaction of Ex4 Analogues 1 with Rhamnolipid Derivatives. Ex4-Rhamnolipid conjugates **5–7** were synthesized via an SrtA-mediated ligation reaction, following the protocol described in our previous research.[27] Ex4 analogue **1** (1.0 μmol, 1.0 equiv) was mixed with rhamnolipid derivatives **2–4** (10 μmol, 10 equiv) and dissolved in 5 mL of Tris buffer (50 mM Tris, 100 mM NaCl, 1 mM CaCl2, pH 7.4), which contained sortase A (6 μM). The reaction was incubated overnight at 30 °C with shaking at 250 rpm. Subsequently, the reaction solution was quenched by the addition of a 0.1% aqueous solution of TFA, and the conversion rate of the reaction was monitored by analytical HPLC. Purification was conducted using semipreparative HPLC, followed by concentration under reduced pressure and lyophilization to obtain highly pure Ex4-Rhamnolipid conjugates **5–7**. Analytical HPLC analysis revealed that all conjugates exhibited purities of ≥95%, as determined by UV absorption at 220 nm (Figures S3–S5). The ESI was calculated for $C_{244}H_{384}N_{58}O_{82}S$ (**5**) $[M + 3H]^{3+}$ m/z = 1,824.58, $[M + 4H]^{4+}$ m/z = 1,368.70, and $[M + 5H]^{5+}$ m/z = 1,095.15, yielding results of 1,825.46, 1,369.19, and 1,095.55, respectively; for $C_{248}H_{392}N_{58}O_{84}S$ (**6**) $[M + 3H]^{3+}$ m/z = 1,855.08, $[M + 4H]^{4+}$ m/z = 1,391.56, and $[M + 5H]^{5+}$ m/z = 1,113.45, yielding results of 1,854.80, 1,391.19, and 1,113.37, respectively; for $C_{252}H_{400}N_{58}O_{86}S$ (**7**) $[M + 3H]^{3+}$ m/z = 1,884.45, $[M + 4H]^{4+}$ m/z = 1,413.58, and $[M + 5H]^{5+}$ m/z = 1,131.07, yielding results of 1,884.47, 1,413.46, and 1,130.99, respectively.

***In Vitro* Cellular Assays for GLP-1R Activation Potency.** The agonistic activity of Ex4-Rhamnolipid conjugates for GLP-1R was assessed through functional assays that measured the cAMP response in HEK-293 cells stably expressing human GLP-1R. For preparation, cells in the exponential growth phase were harvested and suspended in DMEM supplemented with 2% FBS. Subsequently, cells were seeded into 96-well microplates at a density of 2×10^4/well. After overnight adherence, cells were pretreated with 500 μM 3-isobutyl-1-methylxanthine (IBMX, J&K Chemical Technology, Beijing, China) for 40 min to prevent cAMP degradation. Subsequently, the cells were incubated with native Ex4 or Ex4-Rhamnolipid conjugates at varying peptide concentrations, with final concentrations ranging from 0.0001 to 1000 nM, for a duration of 20 min. Cells were subsequently lysed using lysis buffer and centrifuged at 600g for 10 min at 2–8 °C to remove cellular debris. The cAMP content in the HEK-293 cells was determined using a competitive cAMP ELISA according to the manufacturer's instructions (R&D Systems Inc., Minneapolis, MN, USA). Each sample was tested in triplicate, and the in vitro potency of each sample was quantified by determining its half-maximal effective concentration (EC_{50}) using GraphPad Prism version 9.5 (San Diego, USA).

Intraperitoneal Glucose Tolerance Test (IPGTT). Seven-week-old male $Lepr^{db}$ mutation db/db mice (C57BL/6 background) were acclimated to handling for 1 week prior to the experiments. Subsequently, the mice were fasted overnight for 18 h and randomized based on body weight (n = 5). They were dosed with either PBS vehicle or test compounds (native Ex4 or Ex4-Rhamnolipid conjugates **5-7** at 25 nmol/kg) via intraperitoneal injection. A glucose solution (1 g/kg s.c.) was administered after a 30 min interval. Blood samples were collected from a tail nick at designated time points (−30 and 0 min before administration of glucose; and 15, 30, 60, 90, and 120 min after administration), followed by measurement of plasma glucose levels using a glucose meter (Sannuo, Changsha, China). The area under the curve (AUC) of the blood glucose profiles was calculated for the period from 0 to 120 min relative to a baseline of 0%. Throughout the entire experimental duration, unless otherwise specified, the mice were housed at the Animal Experimental Center of Jiangnan University. All animal care and experimental procedures were conducted in accordance with the guidelines approved by the Institutional Animal Care and Use Committee of Jiangnan University (JN. No20201230t0150410[380]).

***In Vivo* Pharmacokinetic (PK) Study.** *Immunization of Normal Balb/C Mice to Generate Anti-Rha Antibodies in Serum.* Female Balb/c mice (4-week-old, 15 ± 2 g) were purchased from Shanghai Slac Laboratory Animal Co., Ltd. (Shanghai, China). Upon arrival, the mice were first housed and kept in the Animal Experimental Center of Jiangnan University for 1 week. The Rha-OVA conjugate (2 mg/mL), dissolved in PBS, was mixed with Alum Adjuvant (Cat. No. 77161, Thermo) in equal volumes, followed by adequate shaking on a vortex mixer to prepare the vaccine formulations. Each mouse was inoculated on day 0 by intramuscular (i.m.) injection of 0.1 mL of the above vaccine formulations and enhanced 3 times on days 7, 14, and 21 via

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subcutaneous (s.c.) injection. Blood samples were collected on day 0 prior to the initial immunization and on days 7, 14, 21, and 28, which was 1 week after each immunization, respectively. The blood was centrifuged to obtain antisera, and the anti-Rha serum titers were determined by enzyme-linked immunosorbent assay.

Pharmacokinetic Determination of Ex4-Rhamnolipid Conjugate 5. The above immunized Balb/c mice were randomly divided into 2 groups (n = 3). Mice were weighed before injection. Each group of mice received a single subcutaneous injection of Ex4 and Ex4-Rhamnolipid Conjugate 5 at a dosage of 75 nmol/kg, as already described in previous research.[35] Blood samples were collected from the tail vein at 0.01, 0.41, 0.83, 1.2, 1.6, 2, 4.8, 6.5, 24, and 36 h after injection and stored in polyethylene tubes containing heparin. The blood samples were centrifuged at 4 °C and 3000 rpm for 15 min to extract the plasma for measurement using exendin-4 ELISA kits (Phoenix Pharmaceuticals Inc., Burlingame, CA, USA). The plasma concentrations of each sample as a function of time were plotted by using Prism 9.5 (GraphPad Software Inc.), and their pharmacokinetic parameters were estimated by the one phase exponential decay model.

Hypoglycemic Duration Test. The long-acting antidiabetic effect of native Ex4 and Ex4-Rhamnolipid conjugates was assessed in db/db mice (n = 5) using a previously described methodology. Prior to the experiment, mice with blood glucose values exceeding 20 mmol/L were selected and divided into six groups based on matched nonfasted blood glucose levels. Mice received subcutaneous injections of PBS, semaglutide, and Ex4-Rhamnolipid conjugates, each at a dosage of 25 nmol/kg. Throughout the experiment, food and water were provided *ad libitum*. The second 1−2 μL blood drop was collected from the tail tip incision to determine blood glucose levels at 0, 1, 2, 6, 8, 10, 12, 24, and 36 h using a OneTouch glucometer (Sannuo, Changsha, China).

Chronic *In Vivo* Studies and Histological Analyses. To evaluate the sustained therapeutic effect of conjugate **5**, which exhibited the best performance in the PK study and hypoglycemic duration test, a chronic treatment experiment was conducted on preimmunized db/db mice with high levels of anti-Rha antibody titers. The db/db mice were assigned to four groups (n = 5) based on their fasting glucose levels. Mice in the positive control group received subcutaneous injections of semaglutide (25 nmol/kg, once daily) or native Ex4 (25 nmol/kg, twice daily at 9 AM and 5 PM), while mice in the experimental group received once-daily subcutaneous injections of conjugate **5** (25 nmol/kg, once daily). The HbA1c levels of mice in each group were determined at the end of the chronic experiment using corresponding assay kits on a BS-420 automatic biochemical analyzer (Mindray, Shenzhen, China). Weekly measurements of FBG levels (fasted for 8 h) were obtained by nicking the tail vein using a one-touch glucometer (Sannuo, Changsha, China). Daily records were maintained for food intake, water consumption, and body weight of mice in each group throughout the entire treatment period. At the conclusion of the chronic experiment, an IPGTT was performed to evaluate glucose tolerance. Specifically, after fasting for 18 h, mice were intraperitoneally injected with glucose (0.5 g/kg), and their blood glucose levels were measured at intervals of 0, 15, 30, 60, 90, and 120 min. Following an overnight fast, mice were euthanized, and blood samples were collected by extracting eyeballs using EDTA to prevent clotting. After centrifugation, the plasma levels of alanine aminotransferase (ALT), aspartate aminotransferase (AST), creatinine (Crea), urea, total cholesterol (TC), low-density lipoprotein cholesterol (LDL-C), high-density lipoprotein cholesterol (HDL-C), and triglycerides (TG) were quantified using corresponding assay kits on a BS-420 automatic biochemical analyzer (Mindray, Shenzhen, China). Concurrently, pancreas tissues from each group of mice were collected and fixed in 4% paraformaldehyde for H&E staining. Immunohistochemistry analyses for insulin in pancreas sections were performed using an Expose mouse-specific alkaline phosphatase detection IHC kit with a dilution ratio of 1:1000 (Abcam, Cambridge, UK).

Statistical Analysis. All data were presented as the means ± SEM. The significance analysis was performed using unpaired, ordinary one-way ANOVA with Dunnett's multiple comparisons in GraphPad Prism 9.5. Statistical significance was determined at a threshold of $p < 0.05$.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jmedchem.5c02146.

Analytical data and characterizations of native Ex4, Ex4 analogue **1**, and conjugates **5**−**7**; flow cytometry evaluation of the HEK293-GLP1-R-AcGFP cell construction; standard curve of the cAMP concentration using a competitive cAMP ELISA; standard curve of Exendin-4 concentration using competitive Exendin-4 ELISA kits; evaluation of multiple IPGTT of conjugate **5** in db/db mice; procedure and protocol for expression and purification of SrtA in *E. coli* BL21 (DE3); protocol for preparation of Rha-carrier protein conjugates; protocol for immunization of db/db mice to generate anti-Rha antibodies in serum; ELISA protocol for Rha-antibody titer determination; protocol for HEK293-GLP1-R-AcGFP construction (PDF)

IC50 and other compound properties(CSV)

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Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

AUC, area under the curve; cAMP, cyclic adenosine monophosphate; DNP, dinitrophenyl; EC_{50}, half-maximal effective concentration; Ex4, exendin-4; FBG, fasting blood glucose; FcRn, neonatal Fc receptor; GLP-1, glucagon-like peptide-1; GLP-1R, glucagon-like peptide1 receptor; HbA1c, hemoglobin A1c; HEK293, human embryonic kidney 293; H&E, hematoxylin-eosin; IPGTT, intraperitoneal glucose tolerance tests; MALDITOF-MS, matrix-assisted laser desorption/ionization time-of-flight mass spectrometry; Rha, L-rhamnose; PEG, polyethylene glycol; $T_{1/2}$, half-lives; T2DM, Type 2 diabetes mellitus; SML, sortase A-mediated ligation; SPPS, solid-phase peptide synthesis

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